Name: ______________________ NetID: ______________

Questions 1–3: One will be chosen to be submitted and graded.

Questions 4–5: If you don't have any questions about your homework, try these.

1. Let n be a positive integer and suppose that we have $2n$ real numbers: $a_1, a_2, \ldots a_n$ and $b_1, b_2 \ldots b_n$. Prove the *Cauchy-Schwarz inequality* below using induction on n.

$$\left|\sum_{i=1}^{n} a_i b_i\right| \leq \left(\sum_{i=1}^{n} a_i^2\right)^{\frac{1}{2}} \left(\sum_{i=1}^{n} b_i^2\right)^{\frac{1}{2}}$$

2. Prove that for any $a, b \in \mathbb{R}$ with $a < b$ the open interval (a, b) is equivalent to $(0, 1)$, i.e., $|(a, b)| = |(0, 1)|$.

3. A real number x is said to be algebraic if there exist **integers** $a_0, a_1, \ldots a_n$ such that

$$a_n x^n + a_{n-1} x^{n-1} + \ldots + a_1 x + a_0 = 0,$$

that is, it is a root of a polynomial with integer coefficients. Prove that the set of algebraic numbers is countable.

Hint: You may use the fact (which you might have seen in class or you can prove separately) that if I is a countable set and for each i, A_i is a countable set then $\cup_{i \in I} A_i$ is a countable set.

Hint: You may also use the fact that a polynomial of degree n can have at most n roots.

For Q4–5, suppose that $A, B \subseteq \mathbb{R}$ are nonempty and bounded from above.

4. Prove that if $A \subseteq B$, then $\sup(A) \leq \sup(B)$.

5. Define $A + B = \{a + b \,:\, a \in A, b \in B\}$.

   a) Determine $\{1, 3, 5\} + \{-3, 0, 1\}$.

   b) Prove that $\sup(A + B) = \sup(A) + \sup(B)$.